Examiner only

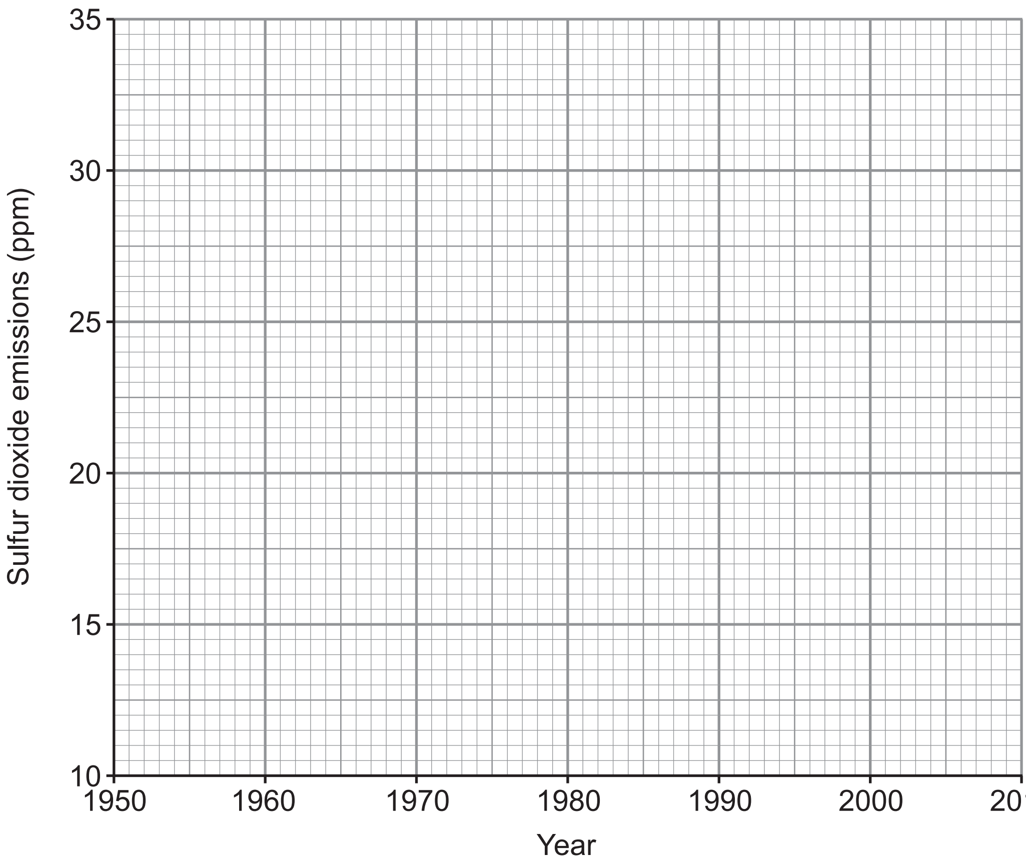
7. Burning fossil fuels containing sulfur causes sulfur dioxide, SO_2, to be released into the atmosphere.

The table shows sulfur dioxide emissions in the UK between 1950 and 2010.

Year	Sulfur dioxide emissions (ppm)
1950	12.0
1960	16.0
1970	21.5
1980	29.5
1990	29.0
2000	24.0
2010	18.5

ppm = parts per million

(a) (i) On the grid plot the sulfur dioxide emissions against the year and draw a suitable line. [3]



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(ii) Describe how sulfur dioxide emissions changed between 1950 and 2010. [2]

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(iii) The UK government introduced a regulation to reduce sulfur dioxide emissions in the 1980s. From your graph, state why it is difficult to decide exactly the year when the regulation came into force. [1]

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(b) Sulfur dioxide can be converted to sulfur and water by reacting it with hydrogen sulfide, H_2S.

Complete and balance the symbol equation for this reaction. [2]

$$SO_2 + \boxed{}\ H_2S \longrightarrow \boxed{}\ S + \boxed{}\ \ldots\ldots\ldots\ldots$$

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